

 kayal2818 / exp-6-DIPT ▾





[Code](#) [Issues](#) [Pull requests](#) [Agents](#) [Actions](#) [Projects](#) [Wiki](#) [Security and quality](#) [Insights](#) [Settings](#)

 **exp-6-DIPT** Public

 Pin  Watch 0 ▾  Fork 0  Star 0 ▾

About

No description, website, or topics provided.

 Readme

 Activity

 0 stars

 0 watching

 0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 1

 kayal2818

 m... ▾

 1 Branch

 0 Tags





 Go to file



Go to file

Add file ➕

Code

⋮

 **kayal2818** Add files via upload 8c968c0 · now 

 EXP 6 DIPT.ipynb Add files via upload now

 README.md Update README with edge detection met... 1 minute ago

README  

exp-6-DIPT

edge-detection-opencv

Aim

To perform edge detection using Sobel, Roberts, Prewitt, Laplacian, and Canny edge detectors.

Software Required

Anaconda – Python 3.7 Jupyter Notebook / VS Code OpenCV (cv2) NumPy Matplotlib

Algorithm

Step 1:

Import all the necessary modules for the program.

Step 2:

Load an image using `cv2.imread()`.

Step 3:

Convert the image to grayscale.

Step 4:

Apply Sobel operator using OpenCV to detect edges.

Step 5:

Apply Prewitt operator using custom kernels.

Step 6:

Apply Roberts operator using custom kernels.

Step 7:

Apply Laplacian operator using OpenCV.

Step 8:

Apply Canny edge detector using OpenCV.

Step 9:

Display all edge-detected images for comparison.

PROGRAM:

Developed By

Name: KAYALVIZHI.V

Register Number; 212225040182

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load image
image = cv2.imread('Fish.jpg') # replace with your path
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# ----- SOBEL -----
```



```
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=5)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=5)
sobel = cv2.magnitude(sobel_x, sobel_y)

# ----- PREWITT -----
prewitt_x = np.array([[1, 0, -1],
                      [1, 0, -1],
                      [1, 0, -1]])

prewitt_y = np.array([[1, 1, 1],
                      [0, 0, 0],
                      [-1, -1, -1]])

prewitt_x_edge = cv2.filter2D(gray, -1, prewitt_x)
prewitt_y_edge = cv2.filter2D(gray, -1, prewitt_y)
prewitt = cv2.magnitude(prewitt_x_edge.astype(np.float32),
                        prewitt_y_edge.astype(np.float32))

# ----- ROBERTS -----
roberts_x = np.array([[1, 0],
                      [0, -1]])

roberts_y = np.array([[0, 1],
                      [-1, 0]])

roberts_x_edge = cv2.filter2D(gray, -1, roberts_x)
roberts_y_edge = cv2.filter2D(gray, -1, roberts_y)
roberts = cv2.magnitude(roberts_x_edge.astype(np.float32),
                        roberts_y_edge.astype(np.float32))

# ----- LAPLACIAN -----
laplacian = cv2.Laplacian(gray, cv2.CV_64F)

# ----- CANNY -----
canny = cv2.Canny(gray, 50, 150)

# ----- DISPLAY -----
```

```
plt.figure(figsize=(12, 10))

plt.subplot(2, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original")
plt.axis("off")

plt.subplot(2, 3, 2)
plt.imshow(sobel, cmap='gray')
plt.title("Sobel")
plt.axis("off")

plt.subplot(2, 3, 3)
plt.imshow(rewitt, cmap='gray')
plt.title("Rewitt")
plt.axis("off")

plt.subplot(2, 3, 4)
plt.imshow(roberts, cmap='gray')
plt.title("Roberts")
plt.axis("off")

plt.subplot(2, 3, 5)
plt.imshow(laplacian, cmap='gray')
plt.title("Laplacian")
plt.axis("off")

plt.subplot(2, 3, 6)
plt.imshow(canny, cmap='gray')
plt.title("Canny")
plt.axis("off")

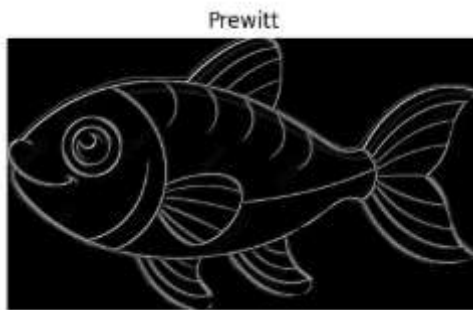
plt.tight_layout()
plt.show()
```

Output

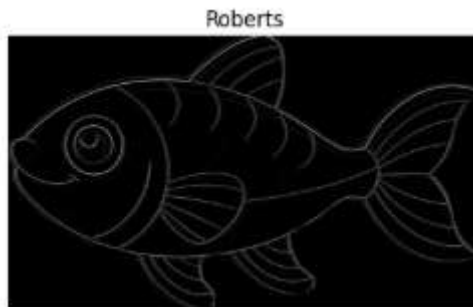
Sobel Edge Detector

<img width="252" height="164" alt="image" src="<https://github.com/user-attachments/assets/aca109c1-42a3-40b3-a987-7d32e9c7cfef>"

Prewitt Edge Detector



Roberts Edge Detector



Laplacian Edge Detector

Laplacian



Canny Edge Detector

Canny

